INTERNATIONAL MASTER IN DATA SCIENCE
ROME BUSINESS SCHOOL

GROUP PROJECT ASSIGNMENT

Build an Image Recognition System: From Image Classification to a Business Application

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Repository / Colab Notebook:
https://colab.research.google.com/drive/12P3E2Hi82A15E270-MuWwS8e4PAUjCqD?usp=sharing

1. The Problem

Plant diseases represent a major threat to global agricultural productivity, causing substantial crop loss and financial insecurity for farmers. Early and accurate disease identification is essential to mitigate damage, prevent widespread infection, and optimize intervention strategies.

In this project, our team engineered an automated computer vision classification system utilizing deep learning to distinguish between healthy and diseased crop leaves. The core objective of the model is to categorize leaf imagery into three target classes: Potato Healthy, Potato Late Blight, and Corn Common Rust. Beyond technical execution, this study explores how image recognition technology can be translated into a practical, decision support business application for agricultural monitoring.

2. The Dataset

The primary dataset deployed in this study is 'PlantVillage Class,' accessed via Roboflow Universe (https://universe.roboflow.com/classification). While the master PlantVillage dataset encompasses 5,966 images across 12 distinct plant and disease categories, our team extracted a focused, task-specific subset of 1,246 images spanning three target classes. The selected working subset was partitioned into training, validation, and test subsets to facilitate model training, hyperparameter tuning, and unbiased evaluation.

3. The Classes

The three selected target categories exhibit distinct visual and pathological characteristics:

• Potato Healthy: Leaves demonstrating uniform green coloration, intact structural margins, and an absence of necrotic lesions or discoloration.
• Potato Late Blight: Leaves exhibiting prominent, dark brown-to-black necrotic patches spreading across larger

areas of the leaf surface.

• Corn Common Rust: Leaves marked by small, raised, reddish-brown pustules or spots distributed across the elongated leaf blade.

Target Class	Training Set	Validation Set	Test Set
Potato Healthy	112	20	20
Potato Late Blight	507	64	63
Corn Common Rust	482	61	60

Table 1: Dataset Partitioning & Class Distribution

4. Dataset Exploration



Figure 1: Sample images showing Potato Healthy, Potato Late Blight, and Corn Common Rust.

Class Imbalance: The training set exhibits noticeable class imbalance. Potato Healthy contains 112 training samples, whereas Potato Late Blight (507) and Corn Common Rust (482) contain over four times as many samples. This imbalance introduces a risk of majority-class prediction bias.



Figure 2: Distribution of training images per class.

The dataset is not perfectly balanced. Potato Healthy has only 112 training images, while Potato Late Blight (507) and Corn Common Rust (482) have significantly more. This imbalance could bias the model toward predicting the majority classes more often, since it sees far fewer healthy potato examples during training. Late Blight and Rust are easy to tell apart from Healthy, but Rust's small spots could be harder to distinguish from early-stage blight.

A few realistic ambiguity points to mention:

- Early-stage Late Blight could look close to a healthy leaf if the disease has only just started.
- Corn Common Rust spots are small and scattered in early stages, this could resemble minor natural leaf blemishes or dirt/dust on the leaf, not necessarily disease
- Lighting and background differences (some leaves photographed on gray backgrounds, some on wood/soil) could confuse a model relying too much on background rather than the leaf itself

Characteristics could help the model distinguish between the classes:

- Color uniformity, Healthy leaves show consistent green color across the whole surface, while diseased leaves show color breaks (brown/dark patches)
- Texture/pattern of damage, Late Blight shows a large, connected necrotic (dead tissue) region; Corn Rust shows many small, separated circular spots a very different spatial pattern even though both are "damage"
- Spot size and density, Rust spots are small and numerous; blight damage is large and concentrated in one area
- Edge sharpness, Blight tends to have a somewhat irregular, spreading edge; rust spots are more discrete/rounded

5. The Model

Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/mobilenet_v2/mobilenet_v2_weights_tf_dim_ordering_tf_kernels_1.0_224_no_top.h5

To prepare raw images for neural network ingestion, all input images are resized to a uniform spatial resolution of 224 x 224 x 3 pixels. Pixel intensity values are scaled to the range [-1, 1] adhering to MobileNetV2 preprocessing standards.

To balance computational efficiency with classification accuracy, our team implemented Transfer Learning using a pretrained MobileNetV2 backbone (pretrained on ImageNet). The convolutional feature extractor was frozen, and a custom classification head was appended consisting of GlobalAveragePooling2D, Dense (128), and Dense (3) with Softmax activation.

mobilenetv2_1.00_224 (Functional)	(None, 7, 7, 1280)	2,257,984
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1280)	0
dense (Dense)	(None, 128)	163,968
dense_1 (Dense)	(None, 3)	387

Total params: 2,422,339 (9.24 MB)
Trainable params: 164,355 (642.01 KB)
Non-trainable params: 2,257,984 (8.61 MB)
We only trained a small classification head on top of a frozen pretrained MobileNetV2, which is efficient and appropriate for a small dataset.

6. Training and Performance

The model was trained over 10 epochs using mini-batch gradient descent. The network achieved 100% training accuracy by Epoch 6 while validation accuracy stabilized around 97.24%. A slight rise in validation loss past Epoch 5 points to mild overfitting.

Epoch	Training Loss	Training Accura
1	0.1961	91.92%
2	0.0488	97.73%
3	0.0208	99.36%
4	0.0160	99.73%
5	0.0132	99.55%
6	0.0066	100.00%
7	0.0053	100.00%
8	0.0030	100.00%
9	0.0023	100.00%
10	0.0020	100.00%

Plot Training Curves:

Validation Loss	Validation Accuracy
0.0821	96.55%
0.0843	97.24%
0.0771	97.24%
0.0819	96.55%
0.0625	97.24%
0.0709	97.24%

0.0702	95.86%

0.0737	97.24%
0.0765	97.24%

0.0732	97.93%

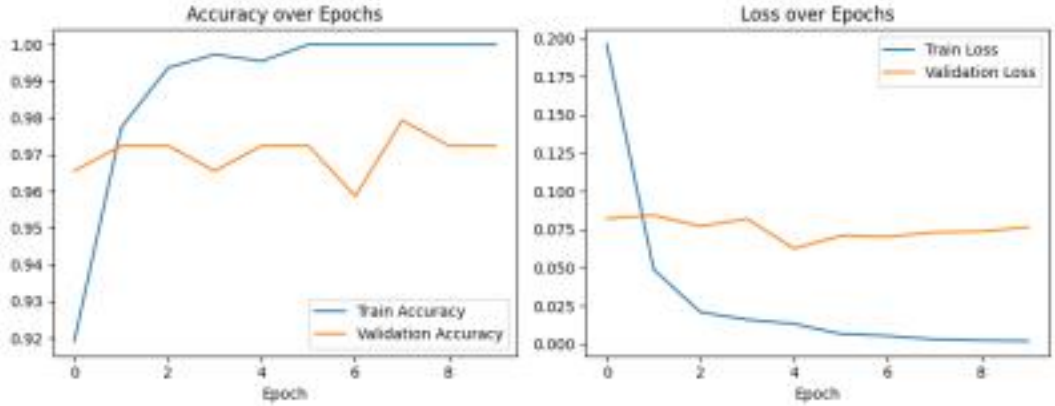


Figure 3: Training Accuracy/Loss vs Validation Accuracy/Loss across 10 Epochs.

The model reaches 100% training accuracy by epoch 6, while validation accuracy plateaus around 97%. Training loss continues to decrease toward zero, but validation loss slightly increases after epoch 4, indicating mild overfitting. This is expected given the frozen pretrained base and relatively small dataset.

7. Prediction Examples & Failure Analysis

Zero cross-species confusion occurred between Corn and Potato classes. All misclassifications occurred exclusively between Potato Healthy and Potato Late Blight due to subtle, localized lesion boundaries in early stage infections.

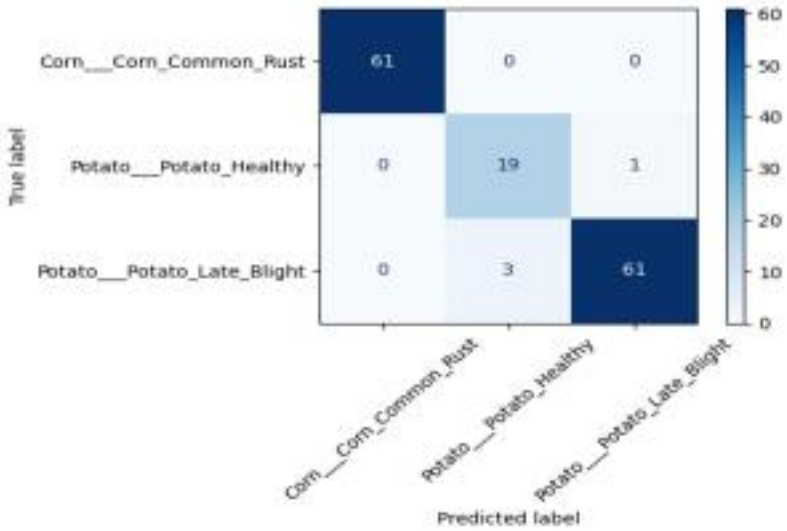


Table 2: Test Set Classification Performance Matrix

True Class	Predicted: Rust	Predicted: Healthy	Predicted: Late Blight	Class Accuracy
Corn Common Rust	61/61	0	0	100.0%
Potato Healthy	0	19/20	1	95.0%
Potato Late Blight	0	3	61/64	95.3%

All misclassifications occurred between Potato Healthy and Potato Late Blight the model never confused either Potato class with Corn Common Rust. This is expected, since Corn leaves are visually very different in shape and texture from Potato leaves, while Healthy and Late Blight Potato leaves share the same overall shape and only differ in localized disease patches likely making early-stage or partially affected leaves harder to distinguish.



Figure 6: Out-of-sample testing on real potato leaves vs. out-of-scope Pothos leaves.

Find and show a genuine misclassification:



The one clear misclassification example shows a Late Blight leaf with only a small brown patch confined to the top edge; the rest of the leaf is visually indistinguishable from a healthy one. This suggests the model relies heavily on the overall proportion of damaged leaf area; when disease is present but minimal, the leaf's dominant visual features still resemble the Healthy class, leading to a wrong prediction. This also reflects a real-world challenge: early-stage disease detection is inherently harder than detecting advanced symptoms.

Model Test

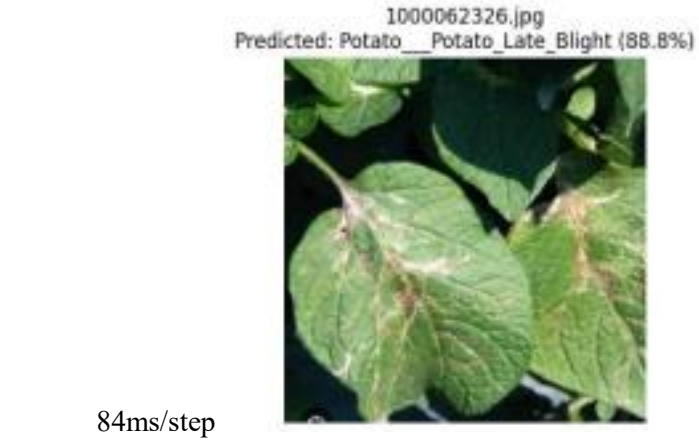
We tested the model on four new images not included in the training dataset: two real potato leaf photos, and two Pothos (a different plant species) leaf photos with unrelated brown spot damage.

1/1 ——————————————————————— **0s**



89ms/step

1/1 ——————————————————————— **0s**



84ms/step

1/1 ——————————————————————— **0s**



116ms/step

1/1 ——————————————————————— **0s** 87ms/step



On the real potato test - success:

On the two real potato leaf photos, the model correctly identified Late Blight with high confidence (98.1% for the classic lesion, 88.8% for the vein-streaking pattern). This confirms the model generalizes reasonably well beyond its original training images, even on an unusual symptom presentation it hadn't seen before.

On the Pothos test - the real insight:

When tested on Pothos leaves; a completely different plant species with brown spot damage unrelated to any disease in our dataset the model still confidently predicted Late Blight (97.6-97.9%). This reveals an important limitation: the model has no concept of 'not applicable' or 'unknown class.' It will always force a prediction into

one of its three trained categories, even for entirely unrelated input. In this case, it appears the model has learned to associate brown/dark patches on a green leaf with Late Blight in general, regardless of the plant species, a shortcut that works on potato leaves but fails when applied to leaves from a different plant.

The potato predictions make sense and reflect genuine model competence. The Pothos predictions, while technically 'wrong,' are explainable: the model appears to associate any brown/dark patch on a green leaf with Late Blight, a pattern that happened to also appear on the unrelated Pothos leaves. This shows the model learned a visual shortcut — 'dark patch = disease' — rather than deeper species-specific features, which is a meaningful limitation to flag.

8. Business Application: Proposed AgriVision Mobile Solution

Problem: Farmers often cannot identify plant diseases early enough, especially without easy access to agricultural experts. Undetected diseases like Late Blight or Common Rust can spread quickly and destroy entire crops, leading to major financial losses.

The User: Small-scale farmers, agricultural cooperatives, or field technicians who inspect crops regularly but may lack specialized plant pathology training.
The Solution (AgriVision Mobile app): A farmer takes a photo of a leaf using a mobile app. The image is sent to our classification model, which predicts whether the plant is healthy, has Late Blight, or has Common Rust. Based on the result, the app displays a recommended action, for example, alerting the farmer to apply fungicide or isolate the affected plants.
The Value Proposition: Early disease detection can prevent large-scale crop loss, reducing the need for costly emergency interventions later. It also removes dependence on expert visits, which may be slow or expensive to arrange, especially in rural areas.

9. Limitations and Risks

- **Inaccurate predictions on early-stage disease**

  Risk: As shown in your own misclassification example, the model struggles when disease symptoms are minimal, a Late Blight leaf with only a small patch was predicted as Healthy.

  Mitigation: Collect and include more training images showing early/mild disease stages, not just clearly advanced cases, so the model learns subtler visual cues.

- **Class imbalance**

  Risk: Potato Healthy had far fewer training images (112) than the other two classes (~500 each), which could bias the model toward predicting the majority classes more often.

  Mitigation: Gather more Healthy images, or use data augmentation (rotation, flipping, brightness changes) specifically on the underrepresented class to balance it artificially.

- **Model cannot recognize out-of-scope or unrelated inputs**

  Risk: When tested on leaves from a completely different plant species (Pothos, not part of the training data), the model still confidently predicted "Potato Late Blight" (97.6–97.9% confidence) rather than

indicating the image didn't match any known class. This shows the model has no way to signal uncertainty or "unknown" — it will always force a prediction into one of its three trained categories, even for irrelevant input.

Mitigation: Add a confidence threshold — if the model's top prediction confidence falls below a certain level, or if input validation confirms the photo doesn't resemble a potato/corn leaf shape, the app could flag the result as "uncertain" rather than presenting a confident (but wrong) answer. Alternatively, train an additional "other/unknown" class using non-target images

- **Overfitting**

Risk: Training accuracy reached 100% while validation accuracy plateaued at ~97%, indicating the model may not generalize perfectly to completely new data.

Mitigation: Use techniques like dropout, data augmentation, or early stopping to reduce overfitting; also t est with a larger, more diverse dataset.

10. Conclusions

This project demonstrated the full pipeline of building an image classification system: from selecting a real-world problem and dataset, through data exploration, model design, training, evaluation, and finally testing the system on new, unseen images.

Using transfer learning with a frozen pretrained MobileNetV2 base, we built a lightweight but effective classifier that distinguishes between Potato Healthy, Potato Late Blight, and Corn Common Rust leaves, reaching approximately 97% validation accuracy. The confusion matrix showed that all classification errors occurred between the two visually similar Potato classes, while Corn Common Rust visually distinct in shape and texture was never confused with either Potato class.

Testing the model on genuinely new images reinforced these findings. On real potato leaf photos, the model correctly identified Late Blight with high confidence, even on an unusual symptom pattern not well represented in training. However, testing on an unrelated plant species (Pothos) revealed an important limitation: the model has no way to recognize input that falls outside its three trained categories, and will confidently force a prediction regardless of relevance. This traced back to a visual shortcut the model appears to have learned associating any dark, irregular patch on a green leaf with Late Blight which works well for its intended task but fails when applied more broadly.

Overall, the model performs well for its intended purpose and demonstrates a real path toward a practical agricultural tool, such as the mobile disease-detection app proposed in this report. At the same time, this project highlights genuine, evidence-based limitations class imbalance, difficulty with early-stage disease symptoms, mild overfitting, and an inability to reject out-of-scope input each of which would need to be addressed before deploying such a system in a real-world setting. These findings reflect not just what the model can do, but an honest understanding of where it falls short and why, which was the core objective of this project.

Notebook: https://colab.research.google.com/drive/12P3E2Hi82A15E270-MuWwS8e4PAUjCqD?usp=sharing